NETFLIX DATA ANALYSIS USING SQL

INSIGHTS INTO CONTENT TRENDS AND PATTERNS

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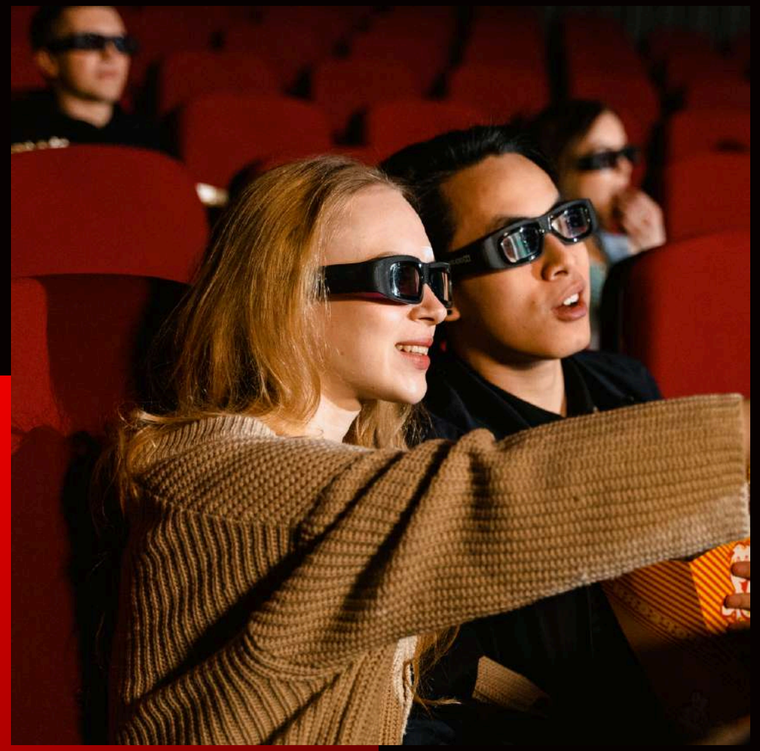


ABOUT PROJECT

Analyzed 8,000+ Netflix titles across 12 attributes to uncover content patterns, regional trends, and key contributors using SQL.

Performed exploratory data analysis to study content trends, ratings, genres, and country-wise distribution.

SKILLS & TOOLS USED

Tools:

SQL (PostgreSQL / pgAdmin)
Microsoft Excel (for data preview & validation)

Technical Skills:

Data Cleaning & Preprocessing
Exploratory Data Analysis (EDA)

SQL Concepts Applied:

Aggregation & Grouping (COUNT, AVG, etc.)
Subqueries & Filtering

DATASET DESCRIPTION

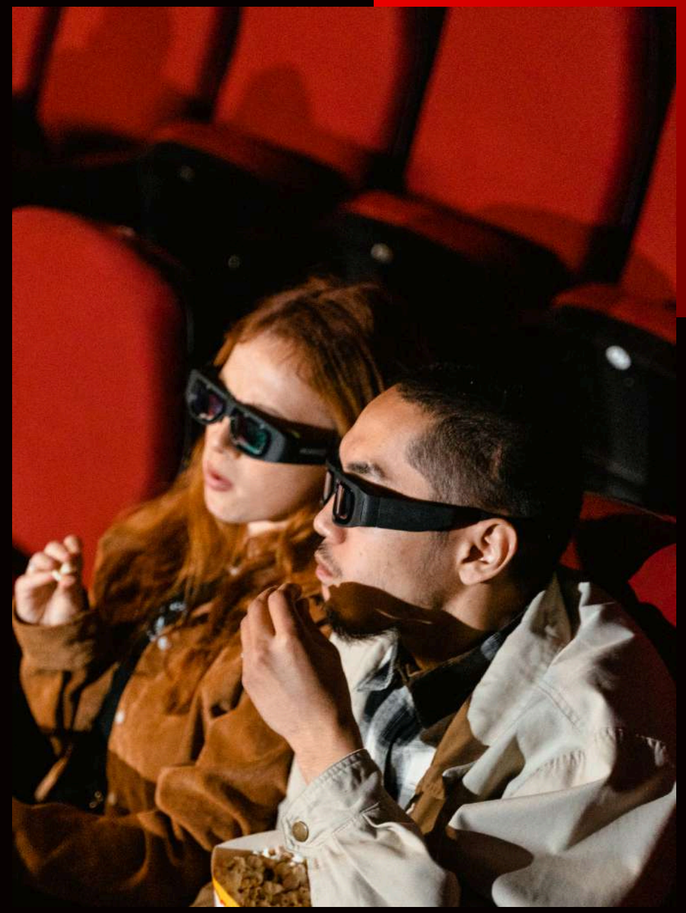
The dataset includes Netflix movies and TV shows with details like title, type, cast, country, release year, rating, and genre.

It is used to analyze content trends, distribution, and key insights across different categories.

CONTENT TYPE ANALYSIS: MOVIES VS TV SHOWS

```sql
SELECT TYPE,
       count(*) AS Total_content
FROM netflix
GROUP BY TYPE;
```

	type character varying (10)	total_content bigint
1	Movie	6131
2	TV Show	2676



INSIGHT:

Netflix's library is heavily skewed toward cinema, with Movies making up approximately 70% of the total content compared to TV Shows.

ANALYSIS OF MOST FREQUENT RATINGS

```sql
SELECT TYPE,
       rating
FROM
  (SELECT TYPE,
          rating,
          count(*),
          rank() over(PARTITION BY TYPE
                      ORDER BY count(*) DESC)AS Ranking
   FROM netflix
   GROUP BY 1,
            2) AS NEW_TABLE
WHERE Ranking = 1;
```

	type character varying (10)	rating character varying (10)
1	Movie	TV-MA
2	TV Show	TV-MA

INSIGHT:
THE MOST COMMON RATING FOR BOTH MOVIES AND TV SHOWS IS TV-MA, INDICATING THAT NETFLIX'S CONTENT STRATEGY IS PRIMARILY FOCUSED ON ADULT AUDIENCES.

MOVIES RELEASED IN A SPECIFIC YEAR (2020)

```
SELECT *
FROM netflix;


SELECT title
FROM netflix
WHERE TYPE = 'Movie'
  AND release_year=2020;
```

	title character varying (150)
1	Dick Johnson Is Dead
2	Europe's Most Dangerous Man: Otto Skorzeny in Spain
3	Tughlaq Durbar

Total rows: 517

INSIGHT:

In 2020, Netflix released 517 movies, highlighting a massive content output despite the global production challenges of that year.

TOP 5 COUNTRIES WITH THE MOST NETFLIX CONTENT

```
SELECT unnest(string_to_array(country, ',')) AS Total_country,
       count(show_id) AS Total_content
FROM netflix
GROUP BY country
ORDER BY Total_content DESC
LIMIT 5;
```

	total_country text	total_content bigint
1	United States	2818
2	India	972
3	United Kingdo...	419
4	Japan	245
5	South Korea	199

INSIGHT:

The United States and India are Netflix's primary content hubs, with the US producing nearly triple the content of India, which holds a solid second place.

LONGEST MOVIE ON NETFLIX

```
SELECT title, duration
FROM netflix
WHERE type = 'Movie'and
duration is not null
ORDER BY CAST(SPLIT_PART(duration, ' ', 1) AS INT) DESC
LIMIT 1;
```

	title character varying (150)	duration character varying (15)
1	Black Mirror: Bandersnatch	312 min

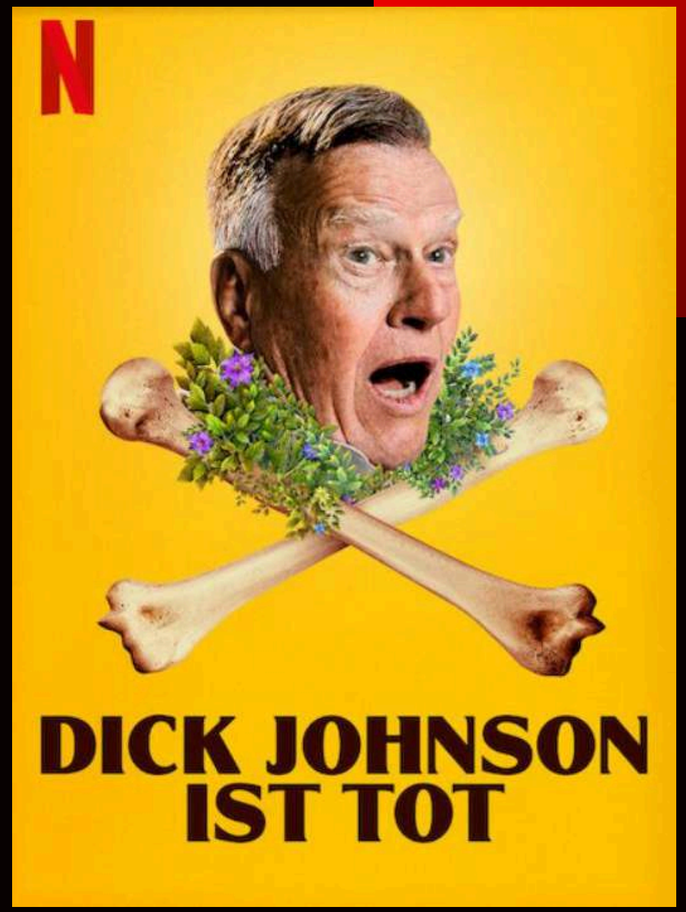
INSIGHT:
Black Mirror: Bandersnatch is the longest movie at 312 minutes, a length driven by its interactive, branching-path format.

CONTENT ADDED IN THE LAST 5 YEARS

```sql
SELECT title,
       to_date(date_added, 'month DD,YYYY')
FROM netflix
WHERE to_date(date_added, 'month DD,YYYY') >= CURRENT_DATE - interval '5 years';
```

	title character varying (150)	to_date date
1	Dick Johnson Is Dead	2021-09-25
2	Blood & Water	2021-09-24

Total rows: 1153



INSIGHT:
This dataset reveals that Netflix added 1,153 titles between 2016 and September 25, 2021, showcasing a rapid library expansion leading up to that cutoff date.

MOVIES & TV SHOWS DIRECTED BY ROHIT SHETTY

```sql
SELECT title,
       director
FROM netflix
WHERE director like '%Rohit Shetty%';
```

	title character varying (150)	director character varying (208)
1	Chennai Express	Rohit Shetty
2	Singham	Rohit Shetty
3	Golmaal Returns	Rohit Shetty
4	Dilwale	Rohit Shetty
5	Golmaal: Fun Unlimited	Rohit Shetty

INSIGHT:

The presence of blockbuster films like Singham and Chennai Express indicates that Netflix prioritizes high-octane, commercial Indian cinema to appeal to the broad Bollywood-loving demographic.

TV SHOWS WITH MORE THAN 5 SEASONS

```sql
SELECT title,
       SPLIT_PART(duration, ' ', 1) AS seasons
FROM netflix
WHERE TYPE = 'TV Show'
  AND SPLIT_PART(duration, ' ', 1)::numeric > 5;
```

	title character varying (150)	seasons text
1	The Great British Baking Show	9
2	Nailed It	6

Total rows: 99

INSIGHT:

The fact that very few TV shows in this dataset (less than 5%) have more than 5 seasons highlights Netflix's "short-lifecycle" strategy, where they prioritize new, 1-3 season limited series over long-running legacy shows to keep the library feeling fresh.

CONTENT DISTRIBUTION BY GENRE

```
SELECT count(show_id) AS Total_content,
       unnest(STRING_TO_ARRAY(listed_in, ',')) AS Total_genre
FROM netflix
GROUP BY Total_genre;
```

	total_content bigint	total_genre text
1	774	International TV Shows
2	9	Stand-Up Comedy
3	172	Spanish-Language TV Shows

Total rows: 73

INSIGHT:
From the 73 rows of distinct categories in your results, it's clear that Netflix’s production strategy isn't just about making "movies"—it's about hyper-specialization

TOP 5 YEARS WITH HIGHEST CONTENT RELEASES IN INDIA ON NETFLIX

```
SELECT count(show_id) AS Total_content,
       EXTRACT(YEAR
               FROM TO_DATE(date_added, 'Month DD,YYYY'))AS released_year
FROM netflix
WHERE country='India'
GROUP BY released_year
ORDER BY Total_content DESC
LIMIT 5;
```

	total_content bigint	released_year numeric
1	333	2018
2	203	2019
3	189	2020
4	142	2017
5	95	2021

INSIGHT:

Netflix content in India peaked in 2018, followed by a gradual decline in subsequent years.

DOCUMENTARY MOVIES ON NETFLIX

```sql
SELECT title,
       TYPE,
       listed_in
FROM netflix
WHERE TYPE= 'Movie'
  AND listed_in like '%Documentaries%'
```

	title character varying (150)	type character varying (10)	listed_in character varying (100)
1	Dick Johnson Is Dead	Movie	Documentaries
2	Europe's Most Dangerous Man: Otto Skorzeny in Spain	Movie	Documentaries, International Movies

Total rows: 869

INSIGHT:

Netflix hosts a strong collection of 869 documentary movies, reflecting high demand for informative content.

ANALYSIS OF MISSING DIRECTOR INFORMATION

```
SELECT title,
       TYPE
FROM netflix
WHERE director IS NULL;
```

	title character varying (150)	type character varying (10)
1	Blood & Water	TV Show
2	Jailbirds New Orleans	TV Show
3	Kota Factory	TV Show

Total rows: 2634

INSIGHT:

Of the 2,634 titles without a director, the examples shown are all TV shows, suggesting that many of these missing values are naturally due to the way series are structured.

MOVIES FEATURING SALMAN KHAN IN THE LAST 10 YEARS

```
SELECT title,
       casts,
       to_date(date_added, 'month DD,YYYY')
FROM netflix
WHERE to_date(date_added, 'month DD,YYYY') >= CURRENT_DATE - interval '10 years'
  AND casts ilike '%salman khan%';
```

	title character varying (150)	casts character varying (1000)	to_date date
1	Hello Brother	Salman Khan, Arbaaz Khan, Rani Mukerji, Johny Le...	2021-01-04
2	Bodyguard	Salman Khan, Kareena Kapoor, Raj Babbar, Asrani, ...	2020-11-19

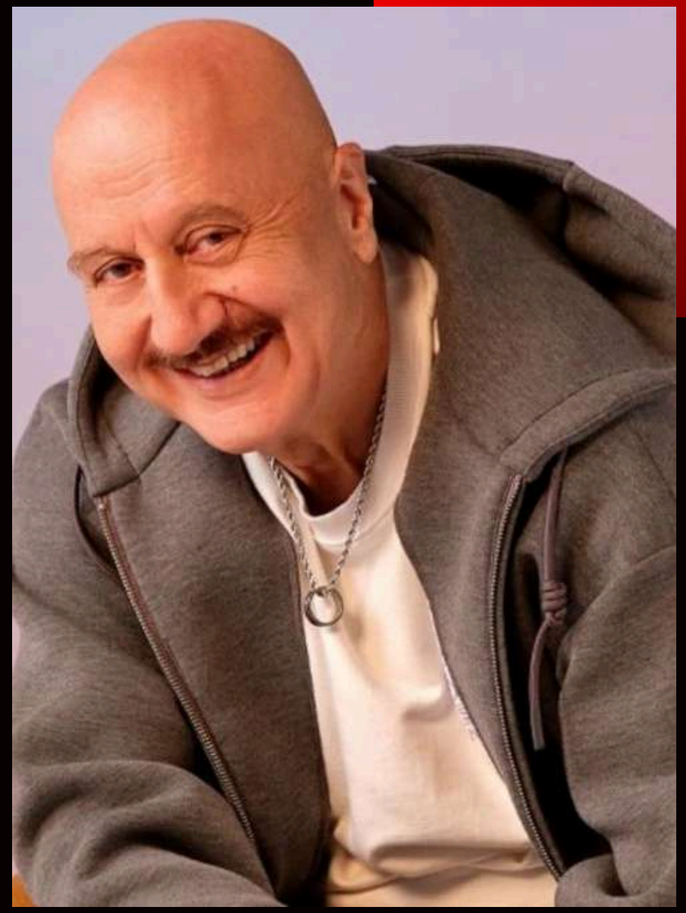
Total rows: 20

INSIGHT:
The query identified 20 Netflix titles featuring Salman Khan that were added in the last decade, with the most recent additions being Hello Brother (Jan 2021) and Bodyguard (Nov 2020).

TOP 10 MOST FREQUENT ACTORS IN INDIAN MOVIES

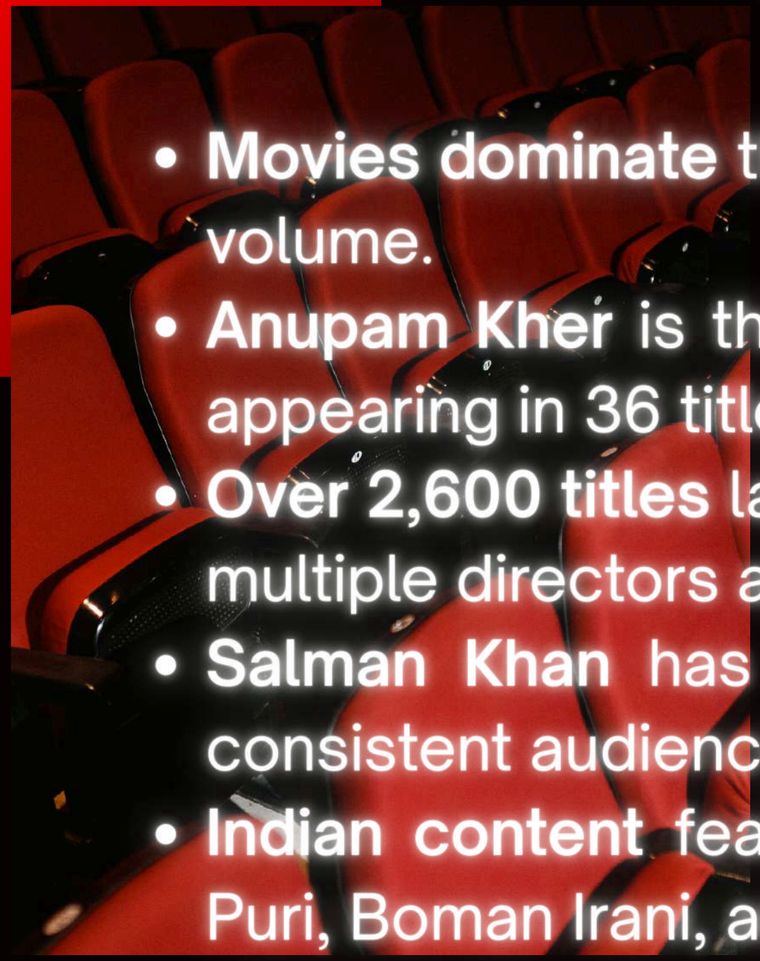
```sql
SELECT unnest(STRING_TO_ARRAY(casts, ',')) AS actor,
       count(show_id) AS number_of_films
FROM netflix
WHERE country ilike '%india%'
GROUP BY actor
ORDER BY number_of_films DESC
LIMIT 10;
```

	actor text	number_of_films bigint
1	Anupam Kher	36
2	Om Puri	26
3	Boman Irani	25
4	Paresh Rawal	25
5	Shah Rukh Khan	25
6	Akshay Kumar	23
7	Naseeruddin Shah	20
8	Amitabh Bachchan	20
9	Kareena Kapoor	20
10	Asrani	17



INSIGHT:

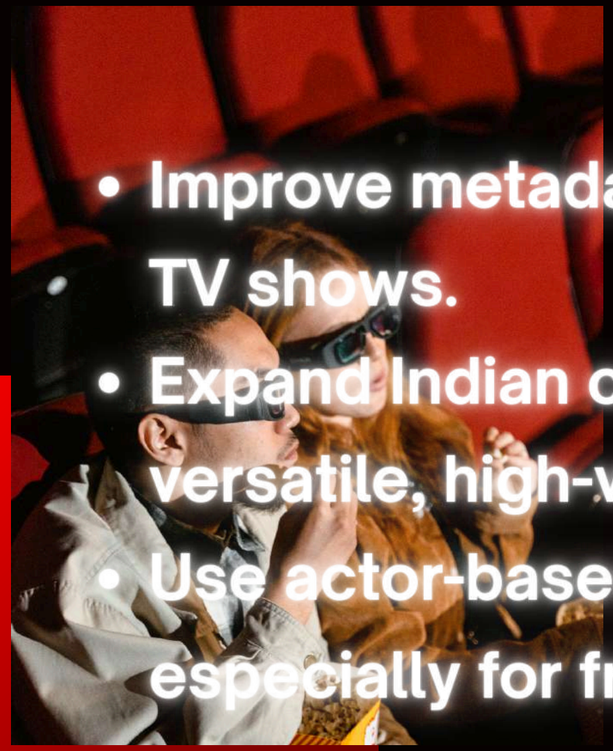
Among Indian content on Netflix, Anupam Kher appears in the most titles (36 films), followed by Om Puri (26), while Shah Rukh Khan, Akshay Kumar, and Amitabh Bachchan also rank in the top 10 most frequent actors.



Five Key Conclusions

- **Movies dominate** the platform, but TV shows are growing rapidly in volume.
- **Anupam Kher** is the most frequent actor in Indian Netflix content, appearing in 36 titles.
- **Over 2,600 titles** lack director information, mostly TV shows where multiple directors are common.
- **Salman Khan** has 20 titles added in the last decade, showing consistent audience demand.
- **Indian content** features a deep roster of veteran actors like Om Puri, Boman Irani, and Shah Rukh Khan.

Suggestions



- Improve metadata quality by standardizing director attribution for TV shows.
- Expand Indian content library given the strong presence of versatile, high-volume actors.
- Use actor-based recommendations to drive engagement, especially for frequently appearing talent.

THANK YOU